

Department of Computer Applications

II MCA – Java Programming Lab – Handson Worksheet 5

Jagged Arrays

Lab 1 – Student Marks Analyzer

Digital Story: Students take different subjects.

Problem: Store marks using jagged array and compute totals/averages.

Sample Input:

2

3

80 75 90

2

60 70

Sample Output:

Student1 Total=245 Avg=81.67

Student2 Total=130 Avg=65.00

Test Cases: single student, zero subjects, many students.

Test Cases

Case	Input	Expected
1	1 student, 1 mark	Same mark as avg
2	multiple students, different sizes	correct totals
3	0 subjects	handle gracefully

Skeleton Java Code:

```
import java.util.*;
class StudentMarks{
    public static void main(String[] a){
        int[][] marks;
        // allocate, input, calculate
    }
}
```

Viva Questions:

1. Why jagged array? 2. Memory saving? 3. How stored internally?

Lab 2 – Theatre Booking

Digital Story: Screens have different seats.

Problem: Track bookings and count available seats.

Sample Input: 2 screens (3,5 seats).

Sample Output: availability count.

Test Cases: all free/all booked/invalid.

Skeleton Java Code:

```
class TheatreBooking{
    public static void main(String[] a){
        int[][] seats;
    }
}
```

Viva Questions:

1. Rectangular vs jagged? 2. Space efficiency? 3. Use cases?

Lab 3 – Graph Adjacency List

Digital Story: Different connections between buildings.

Problem: Store neighbors using jagged array.

Sample: 0->1 2 etc.

Test Cases: disconnected, single node.

Skeleton Java Code:

```
class GraphAdjList{
    public static void main(String[] a){
        int[][] graph;
    }
}
```

Viva Questions:

1. Matrix vs list? 2. Memory? 3. Algorithms using it?

Lab 4 – Sentence Word Counter

Digital Story: Sentences have variable words.

Problem: Count words using String jagged array.

Sample Output: word counts.

Test Cases: empty, single, many.

Skeleton Java Code:

```
class SentenceWords{
    public static void main(String[] a){
```

```
String[][] words;  
}  
}
```

Viva Questions:

1. Why String[][]? 2. split() returns? 3. Longest row logic?

Lab 5 – Store Sales Analyzer

Digital Story: Stores sell different products.

Problem: Calculate totals and find highest.

Sample: totals per store.

Test Cases: zero sales, large values.

Skeleton Java Code:

```
class StoreSales{  
    public static void main(String[] a){  
        int[][] sales;  
    }  
}
```

Viva Questions:

1. Dynamic rows? 2. length vs length()? 3. 3D jagged possible?